Capstone Project 2026

CephFS + SMB High Availability + Virtual Office Network Reference Architecture

Program: IT Systems Management & Security (2-Year Diploma)

Industry Partner: 45Drives

Institution Partner: NSCC Waterfront

Project Duration: Capstone Term (12 weeks)

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1. Project Overview

This capstone project is a full end-to-end enterprise infrastructure build that mirrors real-world deployments supported by 45Drives. Students will design, deploy, secure, and document a **production-grade Ceph storage cluster** and integrate it into a **fully virtualized office network** with identity, security, redundancy, and best practices applied throughout the stack.

Unlike previous research-only projects, this year's capstone focuses on **hands-on implementation of technologies actively used in production by 45Drives**. The goal is twofold:

1. Reinforce and demonstrate the skills learned over the last two years
2. Prepare standout students to step directly into 45Drives work terms and junior engineering roles with practical, job-ready experience

Students will work in teams and will be treated as a small infrastructure engineering group delivering a real solution.

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2. High-Level Architecture

Students will build an enterprise storage and network environment consisting of:

- A **4-node Ceph storage cluster**
- A **2-node SMB gateway cluster with CTDB for HA**
- A **10 GbE physical switching fabric**
- A **compute server running EVE-NG** to simulate a full corporate office

- A **Windows Active Directory domain** providing centralized identity
- A **segmented VLAN-based network with firewall enforcement**

The virtual office network will connect directly to the physical Ceph environment, allowing Windows clients to mount and use SMB shares backed by CephFS exactly as they would in a real company.

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3. Hardware Provided

3.1 Physical Infrastructure

Storage Nodes (x4)

- Linux servers (Rocky Linux 9)
- Multiple data disks per node
- Dedicated system disks
- 10 GbE network interfaces

Gateway Nodes (x2)

- Linux servers (Rocky Linux 9)
- Dual 10 GbE network interfaces
- Used for SMB + CTDB high availability

Network

- 1 × 10 GbE managed switch
- VLAN capable
- LACP support

Compute

- 1 × Compute server
- Runs EVE-NG
- Hosts all virtual office infrastructure

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4. Learning Objectives

By the end of the project, students will demonstrate:

- Enterprise Linux administration
- Distributed storage design and troubleshooting

- High availability services (Ceph, CTDB, SMB)
- Network design, segmentation, and firewalling
- Identity management with Active Directory
- Secure system hardening
- Professional documentation and diagrams

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5. Phase 1 – Research & Design

5.1 Ceph Fundamentals

Students will research and document:

- Ceph architecture (MON, MGR, OSD, MDS)
- Replication vs EC (theory only)
- CRUSH placement basics
- Failure domains and resiliency
- CephFS architecture

Deliverable: **Ceph Design Document**

5.2 Network Design

Students will design:

- VLAN layout for office departments
- Management network
- Storage network (layer 2 non-routed Ceph network)
- Client network (Ceph front-end network that will route to client VLANs)
- Firewall boundaries

Deliverable: **Network Diagram (logical + physical)**

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6. Phase 2 – Ceph Cluster Deployment

6.1 Base OS Configuration

- Install Rocky Linux 9 on all nodes
- Hostname, DNS, time sync
- Disable root SSH
- Configure sudo users

- Apply firewall rules

6.2 Network Configuration

- Create bonded interfaces (active-backup is sufficient)
- Validate MTU consistency
- Test redundancy

6.3 Ceph Installation

- Install Ceph using upstream documentation + 45Drives documentation
- Deploy MON, MGR, OSDs
- Validate cluster health
- Document cluster layout

6.4 CephFS Deployment

- Create metadata and data pools
- Deploy MDS daemons
- Mount CephFS on test clients

Deliverable: **Ceph Build & Validation Report**

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7. Phase 3 – SMB Gateway Cluster

7.1 CTDB High Availability

- Configure 2-node CTDB cluster
- Floating IPs
- Health checks
- Failover testing

7.2 Samba Integration

- Serve CephFS via SMB
- Test failover during active IO

Deliverable: **HA SMB Architecture Document**

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8. Phase 4 – Virtual Office Network (Weeks 9–11)

8.1 EVE-NG Deployment

Students will build a virtual enterprise network including:

- Core router
- Access switches
- Firewall
- VLAN segmentation
- Inter-VLAN routing based on security best-practices

8.2 Windows Infrastructure

- Deploy Windows Server
- Configure Active Directory
- Create users, groups, OUs
- Join all PCs to domain
- Join gateways to Active Directory
- Use AD users/groups for permissions

8.3 Client Integration

- Deploy multiple Windows PCs
- Map SMB shares from Ceph
- Enforce access controls via AD

Deliverable: **Virtual Office Network Demonstration**

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9. Phase 5 – Security Hardening

Students will implement:

- firewalld lockdown (least privilege)
- SSH key-based authentication
- No root login
- Service isolation
- Network segmentation enforcement
- Password policies

Deliverable: **Security Hardening Checklist & Evidence**

10. Phase 6 – Documentation & Presentation

Final Deliverables

1. **Final Architecture Whitepaper**
2. **Network Diagrams (physical + logical)**
3. **Ceph Deployment Guide**
4. **SMB + CTDB HA Guide**
5. **Security Implementation Report**
6. **Recorded Demo or Live Presentation**

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11. Bonus / Advanced Challenges (Optional)

- Snapshot strategy for CephFS
- Simulated failure testing (node down, disk failure, link loss)

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12. Industry Alignment

This project directly mirrors real deployments built and supported by 45Drives.

This capstone is designed to create **job-ready graduates**, not just project reports.
